SereneStat: A Handheld Potentiostat for SWV and Demonstration with a Cortisol Aptasensor

Chengyu Yang, A. Nisha Naeem and Sara S. Ghoreishizadeh
Department of Electrical and Electronic Engineering, Imperial College London, London, U.K.
{chengyu.yang.25, a.naeem, s.ghoreishi}@imperial.ac.uk

***Abstract*—SereneStat is a low-noise, standalone handheld and reconfigurable potentiostat capable of carrying out complete square-wave voltammetry (SWV) measurements. This includes SWV waveform generation and sensor current readout followed by peak current extraction and data display and local storage. Stored records can be retrieved via Bluetooth as well for offline processing. The device operates from a 190 mAh, 3.7 V battery with an estimated battery life-time of 41 days. The PCB system incorporates driven guarding around the sensitive analogue traces, leading to a measured input-referred current noise of 2.4 rms pA. The device was successfully used to measure cortisol with an aptamer-based sensor developed in house.**

***Index Terms*—potentiostat, square-wave voltammetry, low-noise instrumentation, peak extraction, cortisol aptasensor.**

I. Introduction and Related Work

Electrochemical biosensors offer a compact route to biomarker measurement and are increasingly used for point-of-care and repeated testing [1]. These sensors typically produce an electrical current proportional to the concentration of the target molecule when a specified excitation voltage is applied. A portable, handheld electronic instrument, commonly referred to as a potentiostat, is required to excite the sensor, measure its current response, and display the results, as conceptually illustrated in Fig. 1. For the intended repeated point-of-care workflow illustrated in Fig. 1, key requirements are compact form, long battery life, and autonomy—where the device displays immediate measurement independent of other devices (e.g. smartphone or computer).

Square-wave voltammetry (SWV) is a popular electrochemical sensing technique due to its ability to reduce the effect of capacitive current, a.k.a. background current, from the measurement. It achieves this by superimposing alternating small voltage pulses on the conventional voltage ramp to excite the sensor. It then samples the sensor current near the end of the forward and reverse pulses, and subtracts the two currents to obtain a net current response [2]. An integrated SWV reader should ideally combine waveform generation, low-noise current acquisition, and extraction of the peak of the net current over the range of applied voltage.

Several platforms [3]–[10] have been developed for SWV; however, they either lack autonomy due to relying on external devices for current peak extraction and data storage or are limited in their noise performance.

In this work, we present a low-noise, standalone handheld SWV potentiostat for point-of-care measurement of biomolecules using electrochemical biosensors. The configuration parameters can be set through a web app through Bluetooth connection to the device. These are then stored on the device's flash memory. Once set up, the device operates autonomously. The peak SWV current is extracted and displayed on the device, while raw curves and measurement parameters are retained for subsequent wireless retrieval. We here characterise the current readout and SWV excitation performances, and demonstrate its use for cortisol sensing using an aptasensor developed in house.

II. System Architecture

A. Hardware Design

Fig. 2(a) shows the hardware system architecture. An AD5941 analogue front end (AFE) integrates a control amplifier, a switch matrix, as well as a high-speed and a low-power transimpedance amplifier (HSTIA/LPTIA) and a programmable-gain amplifier (PGA). The AFE chip also embeds a 16-bit analogue-to-digital converter (ADC), filters, a first-in, first-out (FIFO) buffer, and a sequencer [11]. A microcontroller (nRF52840) controls the AFE over a serial peripheral interface (SPI), processes data, and manages BLE communication [12]. A power management chip (nPM1300) manages a 190 mAh Li-Po battery, supports USB charging, and provides regulated supplies, including a low-dropout regulator (LDO) for the analogue rail.

A 64 MB flash memory (MX25R6435F) [13] provides non-volatile storage for measurement parameters and raw SWV endpoint currents, so the device only needs to be configured once and holds the information at power-down. An organic light-emitting diode (OLED) display, buttons, and electrode connectors complete the user interface.

The four-layer printed circuit board (PCB) separates analogue, digital/radio, and power regions. External feedback resistors set transimpedance gain and current headroom. Buffered guards surround sensitive electrode and amplifier

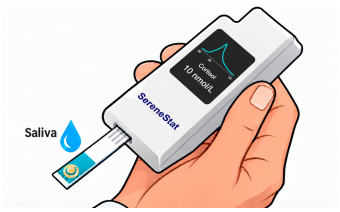
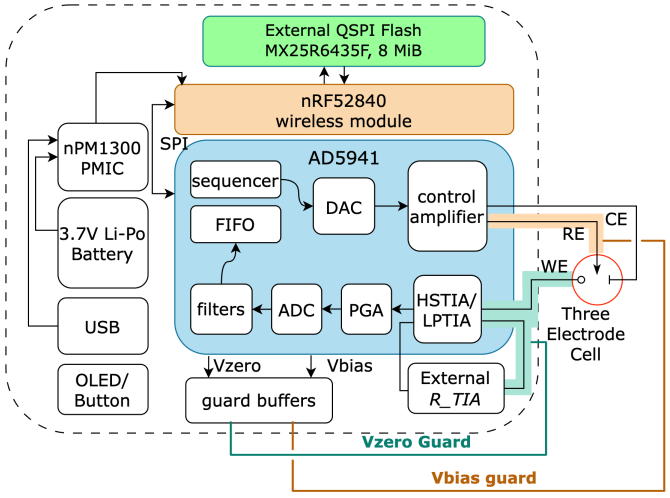
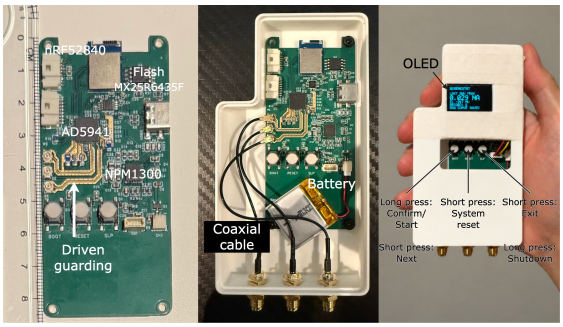


Fig. 1. The envisioned scenario for point-of-care cortisol testing.



(a)



(b)

Fig. 2. (a) System architecture with coloured driven guards; (b) The developed PCB system with sensor contacts, OLED display and enclosure.

nodes to reduce surface leakage [14], [15] and drive the outer conductors of the coaxial interconnects. Fig. 2(b) shows the PCB and packaged prototype. The packaged device measures $11 \times 6 \times 2$ cm.

B. Firmware and Data Flow

The micro-controller firmware runs on the Zephyr real-time operating system (RTOS) [16] and manages user interfaces, BLE communication, AFE operation, and SWV processing. A custom web application configures SWV and peak-processing parameters and the TIA/PGA path, and transfers the related parameter sets over BLE to the device's flash memory. After disconnecting the device from the application, the user can select a stored set of parameters and start an SWV scan using local buttons. The MCU configures the AFE accordingly.

The MCU combines the staircase ramp voltage with the alternating voltage pulses into a sequence of "write and wait" commands for the AFE's 12-bit DAC, then transfers these over SPI to the AFE's 4 kB sequencer SRAM [11]. In addition to the SWV waveform, the DAC also produces a fixed voltage (at half V_{DD}) to bias the reference electrode through the control amplifier.

Sequencer wait instructions set the nominal half-period t_h, giving a frequency $f = 1/(2t_h)$ and a mean scan rate $v = f\Delta E_s$, where ΔE_s is the staircase increment. The AFE

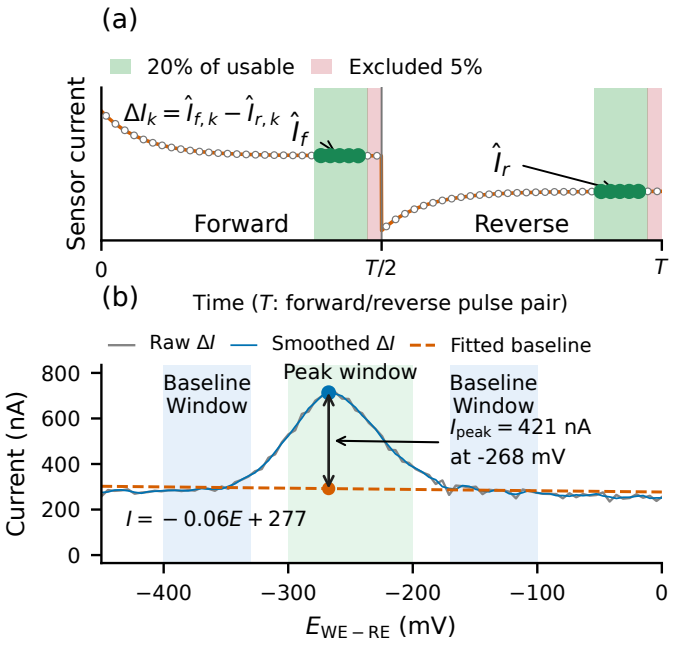


Fig. 3. SWV processing. (a) Current sampling: the final 5% of each half-pulse is excluded, and the last 20% of the remaining interval is averaged. (b) An example of a ΔI_k trace to illustrate the peak detection process.

sequencer controls DAC updates and pulse timing independently of RTOS scheduling.

During a SWV measurement the ADC output passes through cascaded SINC3 and SINC2 digital low-pass filters to suppress high-frequency noise and decimate the data stream. SINC3 provides initial filtering and decimation, while SINC2 allows adjustment of the trade-off between noise reduction and settling time. The filtered samples are transferred through the FIFO to the MCU memory to be processed and stored. The MCU does simple processing to find the net current as shown in Fig. 3(a): it excludes the final 5% of the current measured at each half-pulse and averages the current over the last 20% of the remaining interval (Fig. 3(a)) to obtain the forward and reverse endpoint currents, $I_{f,k}$ and $I_{r,k}$, respectively, where k indexes the pulse number. The net SWV current at each step is calculated as $\Delta I_k = I_{f,k} - I_{r,k}$.

These are then stored for each pulse until a full SWV is completed. Following the peak current extraction routine (explained in the following section), the OLED displays the peak current, while the extracted current peak and raw SWV endpoint currents are stored in the flash memory for further analysis. The user can retrieve stored records through BLE for browser-based display, analysis, or CSV export. A double press of the button toggles BLE advertising.

III. Peak Extraction

The measured net current samples, ΔI_k, first go through a digital five-point, second-order Savitzky–Golay filter (SG5) [19] to smoothen the plot. The smoothed current is

$$\widetilde{I}_k = \frac{1}{35} \sum_{j=-2}^{2} c_j \Delta I_{k+j}, \tag{1}$$

TABLE I
COMPARISON OF BATTERY-POWERED WIRELESS SWV SYSTEMS

Ref.	Current consumption (mA)	Peak detection on MCU?	Local display / data storage?	Input-referred current noise (pA_{RMS})	PCB dimensions (mm)	Target analyte / redox probe
Ainla '18 [3]	4.8	No	No / No	NA	51×34	Ferricyanide; K^+
Wuerstle '21 [4]	1.06	No	No / No	1.2	26 (dia.)	Glucose; alcohol
Lee '22 [5]	NA	No	No / Yes	50 000	40×20	Ferricyanide
Vidotto '23 [6]	≈ 20	No	No / No	NA	NA	Paracetamol
Ait Lahcen '21 [7]	NA	No	No / NR	NA	≈ 37 (dia.)†	HER2
Chien '22 [8]	≈ 3	No	No / No	NA	NA	Kanamycin
Bill '23 [17], [18]	NA	No	No / Yes	NA	51×23	Lactate
Lazaro '25 [9]	NA	Yes	No / No	NA	NA	DNA
Tsyban '25 [10]	NA	Yes	No / No	NA	20×20	Vancomycin
This work	≈ 11.6	Yes	Yes / Yes	2.4	86×40	Cortisol

NA: not available; **NR:** hardware present, function unspecified. **Display/storage** denotes on-device peak-current display/non-volatile raw-SWV storage. †Image-based size estimate.

where the weights c_j, ordered from $j = -2$ to $j = 2$, are $(-3, 12, 17, 12, -3)$.

Then two baseline windows and a peak window are defined as illustrated in Fig. 3(b). The peak window is selected to be $W_P = [-300, -200]$ mV covering the expected methylene-blue redox peak voltage of -250 mV. The baseline windows are selected with a 30 mV clearance from the peak window, at -400 to -330 mV and -170 to -100 mV. A linear model ($B(E) = aE + b$) was then estimated using the least square errors to fit the two regions of ΔI_k over the baseline windows.

The peak current, I_{peak}, is defined as the maximum difference between the smoothed current, $\widetilde{I}_k$, and the fitted line over the peak window, W_P:

$$I_{\text{peak}} = \max_{k:E_k \in W_P} \left[\widetilde{I}_k - (aE_k + b) \right]. \tag{2}$$

IV. ELECTRICAL CHARACTERISATION

A. SWV Voltage Profile

A 1 MΩ resistive load connected the working electrode (WE) to the shorted counter (CE) and reference (RE) electrodes. A myDAQ USB acquisition device (National Instruments) recorded $E_{\text{WE-RE}}$ at 100 kSa/s via a differential analogue input.

The generated SWV voltage waveform measured across the resistive load is shown in Fig. 4. The mean percentage errors between the measured and the expected values of the pulse amplitude, staircase step height, and pulse frequency were -0.1%, -0.07%, and -0.1%, respectively.

B. Input-Referred Noise and Current Range

With all three electrode connections floating, five 60 s battery-powered, radio-quiet recordings were acquired for each of four TIA configurations after 2 s of settling time and 60 s of warm-up time. Cascaded SINC3 and SINC2 filters used oversampling ratios of 5 and 1333, respectively, reducing the ADC output rate from 800 kSa/s to 160 kSa/s and then to 120 Sa/s. The measured ADC output was referred back to the input to find the equivalent input-referred current noise. The noise was calculated as the mean of the standard deviation of the input-referred current across the five recordings.

The results reported in Table II show that the LPTIA configuration (7.5 MΩ, PGA = 9) yields the lowest noise. HSTIA configurations at 330 and 680 kΩ are dominated by the ADC LSB, indicating the input-referred LSB is higher than the actual analogue noise level.

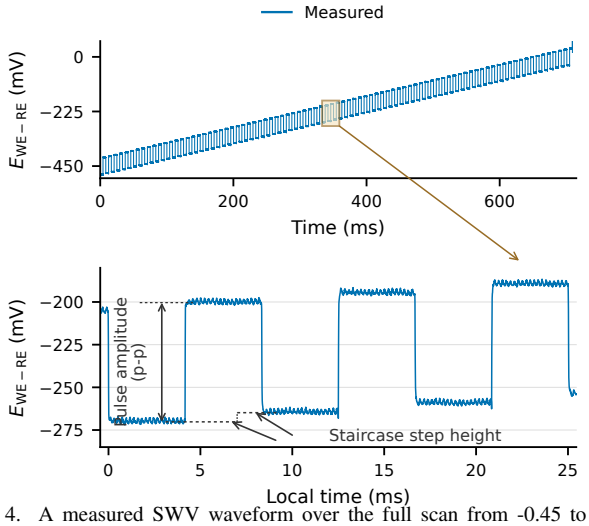


Fig. 4. A measured SWV waveform over the full scan from -0.45 to 0 V, with a magnified view of mid-scan cycles.

Resistive-load tests established the current ranges in Table II, reaching 2.5 μA with an external precision 330 kΩ resistor at PGA = 1.5. Beyond these tested ranges, the AFE supports up to 3 mA with a lower internal feedback resistance, an appropriate PGA gain, and high-current mode [11].

TABLE II
MEASURED HARDWARE PERFORMANCE

TIA type (R_{TIA})	input-referred noise (pA rms)*	Measured DC range (μA) PGA ×1.5	PGA ×9
HSTIA (330 kΩ)	18.9	±2.5	±0.3
HSTIA (680 kΩ)	9.2	±1.2	±0.1
HSTIA (7.5 MΩ)	4.5	±0.1	±0.01
LPTIA (7.5 MΩ)	2.4	±0.1	±0.01

A PGA gain of ×9 was used in noise measurements; feedback capacitance is 4 pF in HSTIA and 220 pF for LPTIA. *The larger of the measured input-referred noise and the input-referred ADC LSB.

C. Power Consumption

The power consumption of the device during various modes of operation is found through the battery current and voltage

log of the power management chip (nPM1300). They are as follows. The mean currents are 10.8 mA during parameter transfer via BLE, 10 mA during the local parameter selection, 10.8 mA during the SWV acquisition, processing and storage, 13.5 mA during OLED result display, and 11.8 mA during the upload of data through BLE.

To find the overall power consumption and battery lifetime, a one-hour window into the expected/typical use scenario is shown in Fig. 5. A short 5 s period is allocated to each of the startup/shutdown times, 10 s to parameter selection, 25 s to SWV acquisition and storage and 10 s to result viewing. With a single 60 s configuration session and an estimated 50 s of upload per day, the total daily energy consumption of the device is 4.6 mAh with 24 periods. A 190 mAh battery therefore provides an estimated 41 days of operation.



Fig. 5. Device modes over a one-hour period. †Ship-mode current uses the PMIC typical value.

V. Electrochemical Measurement of Cortisol

A. Materials

Gold disk working electrodes (1.6 mm diameter) and a silver/silver-chloride (Ag/AgCl) reference electrode were obtained from BASi. The cortisol-specific aptamer (5′-methylene-blue-labelled and 3′-thiol-modified), 6-mercaptohexanol (MCH), cortisol, and magnesium chloride were obtained from Sigma-Aldrich. Tris(2-carboxyethyl)phosphine (TCEP) was obtained from Thermo Scientific. The aptamer sequence is reported in [20].

B. Cortisol aptasensor development

Four identical sensors were developed using the method previously described in [20]. Briefly, the gold electrodes were mechanically polished and electrochemically cleaned following an established protocol [21]. The aptamer was reduced with a 100-fold molar excess of TCEP for 2 h at room temperature in the dark. The electrodes were incubated for 2 h in the dark with 100 nM aptamer in 1× PBS (pH 7.4) containing 2 mM $MgCl_2$. After a 60 s deionised-water rinse, the surfaces were backfilled overnight at 4 °C with 5 mM MCH in 1× PBS containing 2 mM $MgCl_2$. The sensors were rinsed with deionised water and stored in PBS containing 2 mM $MgCl_2$ at 4 °C until use.

C. Measurements

Fig. 6 illustrates the measurement setup. Each sensor was measured in a separate beaker at room temperature using a silver/silver-chloride (Ag/AgCl) reference electrode and a platinum (Pt) counter electrode. All three electrodes were rinsed with deionised water for 60 s before testing. The HSTIA

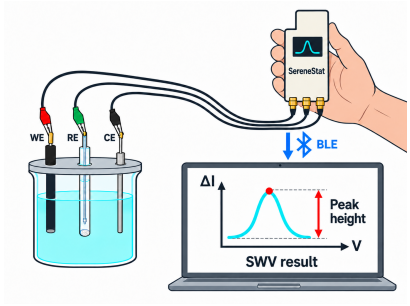


Fig. 6. Schematic of the measurement setup with cortisol aptasensor.

TIA was used with a 330 kΩ resistive load and a PGA gain of 1.5. An SWV voltage waveform was generated as described above with the following parameters: total voltage range of −0.45 to 0 V, pulse frequency of 120 Hz, pulse amplitude of 35 mV, with 5.3 mV staircase increments. Measurements were performed in PBS (pH 7.4) containing 2 mM $MgCl_2$, with a 5 min incubation period before acquisition. At least four blank scans were acquired per sensor in the absence of cortisol, followed by one scan at cortisol concentration of 1 μg/mL.

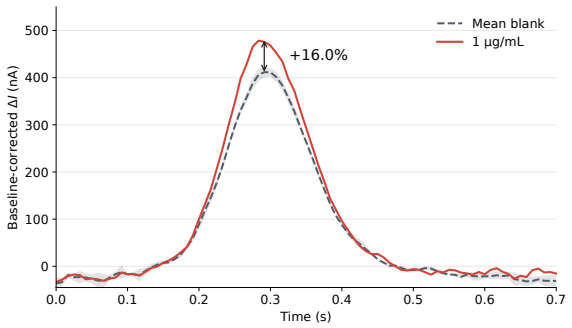


Fig. 7. The smoothed, baseline-corrected measured SWV currents in cortisol-free PBS and at 1 µg/mL cortisol. The dashed curve and shading indicate the blank mean and ±1 SD, respectively.

D. Results

The smoothed, baseline-corrected measured SWV current in a cortisol-free PBS and at 1 μg/mL for one of the sensors is presented in Fig. 7 showing a clear increase in peak current with introduction of cortisol. This was consistent among all four sensors tested with a mean current increase of $11.9 \pm 3.6\%$.

VI. Conclusion

SereneStat supports point-of-care SWV-based measurements without a connected host device, combining locally selectable measurement parameters, on-device peak extraction and display, and raw-data storage for later wireless retrieval. Electrical tests verified SWV waveform generation, a minimum input-referred noise of 2.4 pA rms, and a current range of ±2.5 μA. Cortisol measurements were successfully carried out with SereneStat using a cortisol aptasensor, showing an average increase of 11.9% in SWV peak current in the presence of cortisol.

REFERENCES

[1] A. N. Naeem, S. Guldin, and S. S. Ghoreishizadeh, "Electrochemical sensors for cortisol: A review," *IEEE Sensors Journal*, vol. 24, no. 5, pp. 5746–5758, 2024.

[2] V. Mirceski, R. Gulaboski, M. Lovric, I. Bogeski, R. Kappl, and M. Hoth, "Square-wave voltammetry: A review on the recent progress," *Electroanalysis*, vol. 25, no. 11, pp. 2411–2422, 2013.

[3] A. Ainla, M. P. S. Mousavi, M.-N. Tsaloglou, J. Redston, J. G. Bell, M. T. Fernández-Abedul, and G. M. Whitesides, "Open-source potentiostat for wireless electrochemical detection with smartphones," *Analytical Chemistry*, vol. 90, no. 10, pp. 6240–6246, 2018.

[4] B. L. Wuerstle, "A wireless, reconfigurable, multichannel potentiostat for wearable electrochemical biosensing applications," Master's thesis, University of California San Diego, 2021. [Online]. Available: https://escholarship.org/uc/item/5zd915kj

[5] S. C.-H. Lee and P. J. Burke, "NanoStat: An open source, fully wireless potentiostat," *Electrochimica Acta*, vol. 422, p. 140481, 2022, source code: https://github.com/PeterJBurke/Nanostat (revision 256e5f2).

[6] L. H. B. Vidotto, D. Wachholz Junior, and L. T. Kubota, "A simple and low-cost portable potentiostat with real-time data sharing for wireless electrochemical analyses," *Journal of Electroanalytical Chemistry*, vol. 937, p. 117414, 2023.

[7] A. Ait Lahcen, S. Rauf, A. Aljedaibi, J. I. de Oliveira Filho, T. Beduk, V. Mani, H. N. Alshareef, and K. N. Salama, "Laser-scribed graphene sensor based on gold nanostructures and molecularly imprinted polymers: Application for Her-2 cancer biomarker detection," *Sensors and Actuators B: Chemical*, vol. 347, p. 130556, 2021.

[8] J.-C. Chien, S. W. Baker, K. Gates, J.-W. Seo, A. Arbabian, and H. T. Soh, "Wireless monitoring of small molecules on a freely-moving animal using electrochemical aptamer biosensors," in *2022 IEEE Biomedical Circuits and Systems Conference (BioCAS)*, 2022, pp. 36–39.

[9] A. Lazaro, R. Villarino, D. Girbau, H. Haji-Hashemi, and B. Prieto-Simon, "Low-cost wireless device for DNA sensing using square wave voltammetry," *Chemosensors*, vol. 13, no. 4, p. 119, 2025.

[10] Y. Tsyban, S. S. Shetty, Y. Li, J. I. de Oliveira Filho, J. S. Merzaban, and K. N. Salama, "A portable device for real-time continuous drug monitoring in freely moving small animals," *Biosensors and Bioelectronics*, vol. 287, p. 117704, 2025.

[11] Analog Devices, *AD5940/AD5941 High Precision, Impedance and Electrochemical Front End Data Sheet*, Analog Devices, 2026, rev. G. [Online]. Available: https://www.analog.com/media/en/technical-documentation/data-sheets/ad5940-5941.pdf

[12] Nordic Semiconductor, *nRF52840 Product Specification*, Nordic Semiconductor, accessed: Oct. 9, 2026. [Online]. Available: https://docs.nordicsemi.com/r/bundle/ps_nrf52840/

[13] Macronix International Co., Ltd., *MX25R6435F: Ultra Low Power, 64M-Bit [x1/x2/x4] CMOS MXSMIO (Serial Multi I/O) Flash Memory*, Macronix International Co., Ltd., Aug. 2022, data sheet, Rev. 1.6, Aug. 8. [Online]. Available: https://www.macronix.com/Lists/Datasheet/Attachments/8868/MX25R6435F,%20Wide%20Range,%2064Mb,%20v1.6.pdf

[14] G. Brisebois and J. Wong, "Layout for precision op amps," Analog Devices, technical article, Sep. 2013, accessed: 2026-10-02. [Online]. Available: https://www.analog.com/en/resources/technical-articles/layout-for-precision-op-amps.html

[15] Y. Montes-Cebrián, A. Álvarez-Carulla, G. Ruiz-Vega, J. Colomer-Farrarons, M. Puig-Vidal, E. Baldrich, and P. L. Miribel-Català, "Competitive USB-powered hand-held potentiostat for POC applications: An HRP detection case," *Sensors*, vol. 19, no. 24, p. 5388, 2019.

[16] Zephyr Project, "Zephyr Project Documentation: Introduction," accessed: Oct. 9, 2026. [Online]. Available: https://docs.zephyrproject.org/latest/introduction/index.html

[17] D. Bill, M. Jasper, A. Weltin, G. A. Urban, S. J. Rupitsch, and J. Kieninger, "Electrochemical methods in the cloud: FreiStat, an IoT-enabled embedded potentiostat," *Analytical Chemistry*, vol. 95, no. 35, pp. 13 003–13 009, 2023.

[18] IMTEK-FreiStat, "FreiStat-Framework: Square Wave Voltammetry," 2022, accessed: Oct. 15, 2026. [Online]. Available: https://github.com/IMTEK-FreiStat/FreiStat-Framework/wiki/Square-Wave-Voltammetry

[19] A. Savitzky and M. J. E. Golay, "Smoothing and differentiation of data by simplified least squares procedures," *Analytical Chemistry*, vol. 36, no. 8, pp. 1627–1639, 1964.

[20] N. Naeem, P. Estrela, and S. S. Ghoreishizadeh, "Electrochemical aptamer sensing of cortisol in saliva: mitigating matrix effects via interfacial and signal strategies," *Advanced Biosensors*, vol. 1, p. 100004, 2026.

[21] Y. Xiao, R. Y. Lai, and K. W. Plaxco, "Preparation of electrode-immobilized, redox-modified oligonucleotides for electrochemical DNA and aptamer-based sensing," *Nature Protocols*, vol. 2, no. 11, pp. 2875–2880, 2007.